CohortCert: Shared Evidence and Proof Costs for Cohort Decisions

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***Abstract*—Comparative text analytics must justify every requested decision while controlling expensive predicate evaluations. CohortCert connects shared evidence acquisition to the cost of completing comparisons, equivalence bundles and strict rankings. Signed contrasts combine overlapping records, and established count confidence sequences support true and false decisions. Nested intervals yield a compact completion program. Feasible allocations and independently checked rational lower bounds distinguish available proof opportunities from cost barriers within a specified inference architecture. The offline frontier verifies within-budget proofs in $157/192$ preselected cases; online policies use observed evidence only. Controls retain all $73{,}760$ planned saved-label policy replays, including unresolved and failed requests. DifficultyGreedy improves CivilComments completion over width allocation by 0.78/1.16 percentage points for crowd/model references; GoEmotions differs by two completions among $4{,}640$ units. A scalar reduction explains where the two allocators coincide. Same-path endpoint relaxation proves an additional proof-cost requirement in $30/192$ cases. Forecast optimizes a completion surrogate yet loses to simpler acquisition, separating surrogate planning from execution efficiency. A released-judge request compares seven configured programs, and 23 live query executions account for $69{,}799$ corpus model forwards. The resulting evidence identifies sharing effects and completion costs conditional on the specified labels and architecture.**

***Index Terms*—semantic query processing, finite-population inference, Boolean certificates, evidence acquisition, cohort comparison**

I. Introduction

A complaint-monitoring request may ask whether cohort A has a higher rate than B, whether other rates are practically equivalent, and which cohorts form a strict top-k set. With expensive model predicates, the system must choose what to read and justify the *whole request*: every comparison needs a true/false value, and ranking needs the selected set.

Shared evidence makes this a data-management problem. An overlapping record's contribution depends on both cohort denominators. For example, a record shared by cohorts of sizes 100 and 80 contributes $1/100 - 1/80 = -1/400$ to their rate difference. One record can also advance several decisions. Figure 1 shows the same count bounds proving true and false values while another item remains unresolved.

A request that remains unresolved at its budget presents a second question: could a different allocation have completed it, or does completion require more evidence under the same inference rules? A feasible within-budget allocation demonstrates a proof opportunity. A verified lower bound above budget establishes a cost barrier for those rules; overlapping or missing bounds leave the diagnosis unresolved. CohortCert makes both outcomes measurable through a *conditional proof-cost frontier*.

Semantic engines optimize expensive operators [1]–[3], and Evergreen verifies aggregate and ordinal claims [4]. Boolean evaluation [5]–[7] and joint sequential inference [8]–[10] supply related foundations. CohortCert combines complete comparative contracts with shared finite-population evidence [11] and checkable acquisition-cost bounds.

Three modules implement this workflow: a compiler of signed contrasts and Boolean witnesses, a shared-evidence executor with replaceable allocation policies, and an offline cost evaluator. The evaluator uses realized future paths under fixed inference rules; online Forecast plans with hypothetical paths. Their separation exposes proof opportunities left unused by a policy.

Our contributions are (i) a complete-request contract that supports true and false comparisons, equivalence bundles and strict-set identification using shared evidence; (ii) a compact completion formulation with independently verifiable upper and lower proof-cost bounds; and (iii) aligned controls that identify the effects of signed overlap, cross-decision gain aggregation and allocation. Public code, fixed inputs and analysis artifacts accompany the paper.[1]

[1]https://github.com/yzd2002/CohortCert (snapshot `aae1e7b`); reproduction scope is documented in the repository.

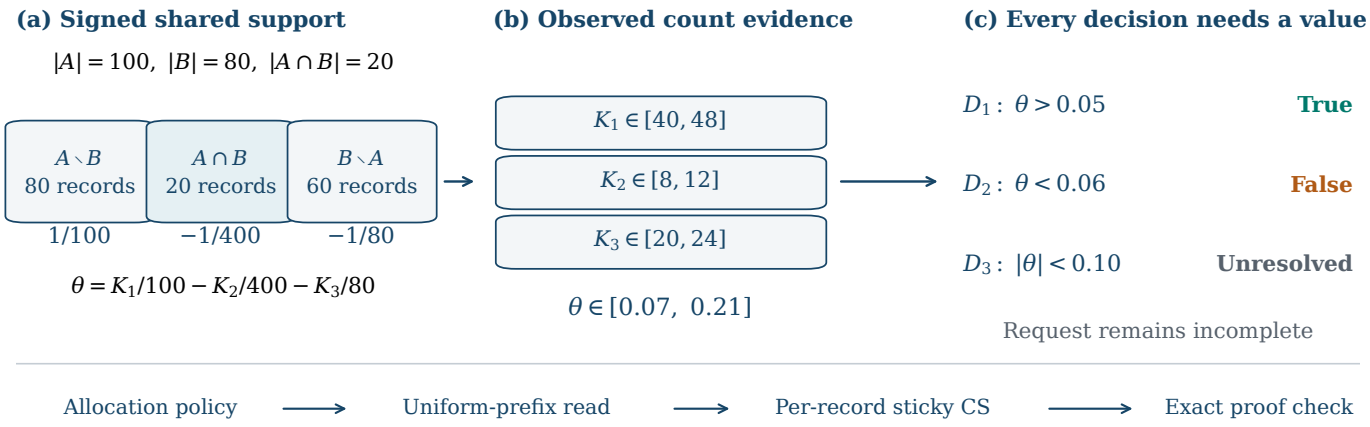


Fig. 1. Shared evidence and complete decisions (illustrative values). K_1, K_2, K_3 count positives in $A \setminus B$, $A \cap B$, $B \setminus A$. Unequal denominators produce a signed overlap coefficient. Count bounds prove D_1 true and D_2 false; D_3 remains unresolved. Different contrasts can also reuse these counts. The allocator chooses reads; intervals justify output.

II. Decision Contracts and Valid Shared Evidence

A request specifies a finite text snapshot, one versioned binary label operator y_i, metadata-defined cohorts G_g with known positive denominators n_g, decision thresholds, a call budget B and risk α. It returns a justified value or unresolved marker for each item, the selected set for a completed ranking, and its acquisition cost. A cohort rate is $\mu_g = \sum_i [i \in G_g] y_i / n_g$. The finite label vector is the inference target; changing crowd to model labels changes that target. Cohort overlap is represented exactly: in $\mu_A - \mu_B$, a shared record has weight $1/n_A - 1/n_B$, which cancels only when the denominators agree.

The compiler produces rational contrasts $\theta = Cy$ and the effective support $I = \{i : \exists j, C_{ji} \neq 0\}$. After observing a pilot P, it groups residual records with identical coefficient columns into cells. Cell h has size N_h and unknown positive count K_h. Each declared decision has a positive witness family

$$D_d = \bigvee_{w \in \mathcal{W}_d} \bigwedge_{j \in J_w} \mathcal{A}_j, \qquad \mathcal{A}_j : \ \beta_j + \sum_h a_{jh} K_h > 0. \quad (1)$$

For a lower comparison $\theta_j > b_j$, choose a positive common-denominator multiplier γ_j, set $a_{jh} = \gamma_j C_{ji}$ for i in cell h, and set $\beta_j = \gamma_j (\sum_{i \in P} C_{ji} y_i - b_j)$. Cohort denominators stay fixed; observed pilot contributions enter β_j. Upper comparisons reverse signs. Equivalence with tolerance $\tau > 0$ is $(\theta + \tau > 0) \wedge (\tau - \theta > 0)$, so boundary equality is false. All atom coefficients and propagated endpoints are integers. Certification is then exact on this lattice: atom j is true if $L_j \geq 1$, false if $U_j \leq 0$, and otherwise unresolved. Equality belongs to the false side of a strict atom.

A true decision needs one witness whose atoms are all certified true. A false decision needs a certified false atom in *every* positive witness:

$$\neg D_d = \bigwedge_{w \in \mathcal{W}_d} \bigvee_{j \in J_w} \neg \mathcal{A}_j. \quad (2)$$

This cover is stored directly; expanding its negative disjunctive normal form is unnecessary. Every item needs its own justified value. For strict top-k, each k-element set S is a witness with atoms $\mu_a - \mu_b > 0$ for all $a \in S, b \notin S$. A completed request returns that S; only positive identification is allowed, including inside mixed requests. No ordering within S is required. Ties may leave strict-set identification unresolved even at census.

A *decision certificate* consists of these asserted atom sides and their count-interval support. This contract uses *atomwise interval proofs*. For example, $\theta > 0$ and $\theta < 0$ are jointly incompatible, but an interval spanning zero refutes neither atom individually. Joint logical reasoning can therefore certify more than our atomwise rules. Our optimization results concern this stated proof system.

A. Budget and Cost

The request fixes a semantic-call cap B and risk α before execution. A call is a submitted record–predicate label attempt, including pilots and failed batches. The objective is correct complete output within B and request-level risk α; unresolved requests stay in the denominator. Within one primary request, each record–predicate label is acquired once and reused by every relevant contrast. Other requests/predicates are charged afresh. Controls share this evidence structure and support filtering. Cohort metadata are given; feature preparation and physical forwards/tokens/time are accounted separately.

B. One Frozen Inference Kernel for All Policies

A charged pilot is sampled uniformly from the support. Its exact contrast contribution is removed, and the residual records are partitioned by identical coefficient columns. The primary revision uses one cell per coefficient signature. A pilot-fitted text proxy fixes a working predictive law; a constant smoothed pilot probability handles a single-class pilot. No unrevealed labels enter the proxy fit. All cells, priors, risks and formal permutations are then frozen.

Let K_h be the unknown positive count of a residual cell of size N_h. A fresh independent uniform permutation supplies each cell's formal observations. After n reads containing s positives, the ordered-path likelihood under candidate K is

$$p_{h,K}(y_{1:n}) = \frac{(K)_s (N_h - K)_{n-s}}{(N_h)_n}. \tag{3}$$

Here $(x)_r = x(x-1)\cdots(x-r+1)$, with $(x)_0 = 1$; the ordered-path probability has no binomial factor. Following sampling-without-replacement confidence sequences [11], invert $q_h(y_{1:n})/p_{h,K}(y_{1:n}) < 1/\alpha_h$, where q_h is a fixed-prior mixture of uniform-Beta and pilot-proxy Beta path laws. With Euler beta function B, this path distribution is

$$q_h = (1-\lambda)\frac{\mathrm{B}(1+s, 1+n-s)}{\mathrm{B}(1,1)} + \lambda \frac{\mathrm{B}(a_h+s, b_h+n-s)}{\mathrm{B}(a_h, b_h)},$$

where $\lambda = 1/2$, proxy strength is 20, and a_h, b_h are frozen pilot-based shapes. Component *prior* weights are fixed; conditional predictive weights update by Bayes' rule. The per-cell risk is uniformly allocated with downward numerical rounding so $\sum_h \alpha_h \le \alpha$.

For the predictable next-label null law p_K and conditional predictive alternative q_t, null-possible histories satisfy

$$\mathbb{E}_K\left[\frac{q_t(Y_t \mid Y_{<t})}{p_K(Y_t \mid Y_{<t})} \,\middle|\, Y_{<t}\right] = \sum_{z: p_K(z \mid Y_{<t})>0} q_t(z \mid Y_{<t}) \le 1. \tag{4}$$

Hence $q_h/p_{h,K}$ is a nonnegative supermartingale even with shrinking null support. Null-impossible prefixes are excluded directly; Ville inversion applies.

At each prefix, accept the integers $K \in [s, s+N_h-n]$ satisfying the ratio inequality. Convex log ratios give a contiguous accepted interval. Starting at $[0, N_h]$, the executor intersects its endpoints with the preceding interval after *every record*, including inside a submitted batch. An empty intersection is a sticky failure: later reads cannot erase it. At exhaustion the exact count is intersected with the preceding interval, rather than replacing an earlier exclusion. The resulting $[\ell_h(n), u_h(n)]$ depends only on that cell's revealed prefix and frozen parameters. For each atom, exact signed propagation gives

$$L_j = \beta_j + \sum_{a_{jh}\ge 0} a_{jh}\ell_h + \sum_{a_{jh}<0} a_{jh}u_h, \tag{5}$$

$$U_j = \beta_j + \sum_{a_{jh}\ge 0} a_{jh}u_h + \sum_{a_{jh}<0} a_{jh}\ell_h. \tag{6}$$

Condition on the snapshot, fixed labels, pilot and frozen partition. Under independent uniform formal permutations and exact likelihood inversion, Ville's inequality and a cell-wise union bound give

$$\Pr\{\forall h, n:\ K_h \in [\ell_h(n), u_h(n)]\} \ge 1-\alpha. \tag{7}$$

On this event all returned atom assertions and decision values are correct. Atoms and witnesses reuse covered counts, requiring no extra multiplicity charge. Coverage is per request, not workload-wide. An adaptive policy can change which cell advances, but receives neither future labels nor unread permutation information. This uses established inference; numerical implementation is audited separately from the exact-arithmetic guarantee.

III. Planning the Shared Certificate Frontier

A. Nested Prefixes Give Additive Covering Gains

Fix a current nonfailed state and a nested continuation for every cell. Locally write $\ell_h(m) = \ell_h(n_h+m)$ and $u_h(m) = u_h(n_h+m)$, so zero denotes the current state and $0 \le m \le N_h - n_h$. For m additional reads, define

$$g^+_{jh}(m) = \begin{cases} a_{jh}(\ell_h(m) - \ell_h(0)), & a_{jh} \ge 0, \\ -a_{jh}(u_h(0) - u_h(m)), & a_{jh} < 0, \end{cases} \tag{8}$$

$$g^-_{jh}(m) = \begin{cases} a_{jh}(u_h(0) - u_h(m)), & a_{jh} \ge 0, \\ -a_{jh}(\ell_h(m) - \ell_h(0)), & a_{jh} < 0. \end{cases} \tag{9}$$

All gains are nonnegative and nondecreasing up to a sticky failure. Direct expansion gives

$$L_j(\mathbf{m}) = L_j(0) + \sum_h g^+_{jh}(m_h), \tag{10}$$

$$U_j(\mathbf{m}) = U_j(0) - \sum_h g^-_{jh}(m_h). \tag{11}$$

Thus true and false atom proofs respectively require gains of at least $d^+_j = \max\{1 - L_j(0), 0\}$ and $d^-_j = \max\{U_j(0), 0\}$. One cell can advance several atoms and decisions in either direction.

B. A Compact Completion Program

For cell h, option k represents m_{hk} additional reads, costs $c_{hk} = m_{hk}$ and supplies their gains. The full frontier allows every legal integer prefix; grid versions retain a subset including zero. Binary x_{hk} selects one option per cell; T_j, F_j select justified atom sides (both may be zero); z_w selects a witness; p_d, r_d choose a true/false decision proof. Unsupported sides cannot be selected. The cost objective is $\min \sum_{h,k} c_{hk}x_{hk}$, subject to

$$\sum_k x_{hk} = 1 \qquad \forall h, \tag{12a}$$

$$\sum_{h,k} g^+_{jh}(m_{hk})x_{hk} \ge d^+_j T_j \qquad \forall j, \tag{12b}$$

$$\sum_{h,k} g^-_{jh}(m_{hk})x_{hk} \ge d^-_j F_j \qquad \forall j, \tag{12c}$$

$$T_j + F_j \le 1 \qquad \forall j, \tag{12d}$$

$$z_w \le T_j \qquad \forall w,\ j \in J_w, \tag{12e}$$

$$\sum_{w \in \mathcal{W}_d} z_w \ge p_d \qquad \forall d, \tag{12f}$$

$$\sum_{j \in J_w} F_j \ge r_d \qquad \forall d,\ w \in \mathcal{W}_d, \tag{12g}$$

$$p_d + r_d = 1 \qquad \forall d. \tag{12h}$$

For a strict ranking-identification decision set $r_d = 0$. With M total prefix options, J atoms, W witnesses, D decisions

and E atom–witness incidences, the program uses $O(M+J+W+D)$ variables and $O(H+J+W+D+E)$ constraints. This representation avoids a negative-DNF expansion; it does not make integer optimization polynomial.

For disjoint pairs, $D = \bigvee_{w=1}^{W}(\mathcal{A}_{w1} \wedge \mathcal{A}_{w2})$ has 2^W negative-DNF terms but only W negative clauses. Primary nonranking decisions have one positive witness, and rankings forbid negative completion. Thus the experiments test shared completion/allocation; the generic representation benefit is separate.

Proposition 1 (exactness). On fixed nonfailed nested paths, a prefix allocation is feasible in (12) if and only if its intervals complete every declared decision under the stated atomwise proof system.

Proof. For a complete allocation, mark its proven atom sides, select one all-true witness for each true decision, and select the false hits for each false decision. The gain identities verify the inequalities. Conversely, the gain constraints establish each selected atom assertion, and the witness constraints implement precisely one complete proof side for each decision. Ranking masks retain identification. □

C. An Auditable Offline Cost Oracle

Only the offline evaluator sees full permutations and realized cell paths. Legal prefixes precede the first sticky inconsistency. Let C^* be the minimum additional-read cost of (12) on all legal prefixes. Every completing policy under these conditions pays at least C^*. An omniscient scheduler can attain that prefix vector in any order by nesting. Mandatory batching/exploration may add cost. Total cost adds prior submissions; our offline study starts after pilot P, giving $|P|+C^*$.

This conditional frontier fixes partition, kernel, risks and proof rules. On noncovering paths, the raw oracle may use a source-wrong atom. Fixing $F_j = 0$ for source-true atoms and $T_j = 0$ for source-false atoms defines the source-valid frontier C^*_{src}. Its selected assertions match the fixed reference, without requiring all cell intervals to cover. Hindsight witness choice is existential: online stopping may choose differently, although no such unattainability is observed on our contracts.

For general uncapped instances, Set Cover reduces to completion: a label-one unit cell represents each distinct set; each element's atom compares its containing-cell mean to an observed zero anchor. Completing all atoms selects a minimum cover. Cohort/cell counts grow in this reduction.

A feasible grid uses actual prefix costs/gains, giving an upper bound. An optimistic grid grants the next endpoint's gains at the preceding endpoint's cost; its exact integer optimum is a lower bound. Rounding that optimum up adds at most its selected bracket widths. LP-based total gaps also include integrality and solver gaps.

We verify lower bounds independently of floating solver claims. For the chosen raw or restricted formulation, let x concatenate prefix and auxiliary variables, encode non-onehot equalities in both inequality directions, and write all such constraints and fixes as $Gx \geq b$. Let $\bar{c} = c - G^\top \eta$. For any rational $\eta \geq 0$, exact weak duality gives

$$\mathcal{L}(\eta) = \eta^\top b + \sum_h \min_{k \in h} \bar{c}_k + \sum_{j \in \mathrm{aux}} \min\{0, \bar{c}_j\} \leq \mathrm{OPT}. \quad (13)$$

This minimizes the Lagrangian over each onehot group and auxiliary $[0,1]$ domain. LP multipliers are converted to nonnegative rationals and checked exactly; an integer-verified allocation supplies the upper bound. These cost-bound certificates differ from online decision certificates. Wide/unavailable bounds remain in the results.

D. Isolating Overlap Endpoint Dependence

Keep the count intervals and zero-net support filtering fixed, but propagate a nonzero intersection contribution through separate left/right endpoints. For one unscaled contrast, this relaxes its signed interval $[L, U]$ to $[L - P, U + P]$, where

$$P = \sum_{h:\, h \subseteq G_A \cap G_B,\ a_h \neq 0} \min\{1/n_A, 1/n_B\}\,(u_h - \ell_h). \quad (14)$$

This identity follows by subtracting the two endpoint expressions. The relaxation leaves the pilot contribution exact. It preserves coverage but may delay either proof direction. Every source-valid relaxed proof is also a signed proof, hence $C^*_{\mathrm{src,signed}} \leq C^*_{\mathrm{src,relaxed}}$. Writing $\underline{C}, \overline{C}$ for total-cost bounds, a checked inequality $\overline{C}_{\mathrm{signed}} < \underline{C}_{\mathrm{relaxed}}$ certifies a positive additional proof-cost barrier, rather than estimating it from point bounds.

The same physical cell retains one paid prefix through two linked appearances; zero-net contributions stay zero. Only endpoint propagation changes, isolating signed overlap rather than all acquisition reuse.

E. A Bounded Online Forecast Planner

Forecast maps current evidence to a hypothetical completion allocation. Define count width $v_h(n) = u_h(n) - \ell_h(n)$ and center $s + (N_h - n)\mathbb{E}_q[p \mid y_{1:n}]$, using the mixture's posterior component weights. Round this total half-up and clip it to the current interval. Prefix zero keeps that interval; for $0 < n < N_h$ and $0 < m \leq N_h - n$, predicted width is

$$\widetilde{v}_h(n+m) = v_h(n)\sqrt{\frac{n(N_h - n - m)}{(n+m)(N_h - n)}}. \quad (15)$$

Inward integer rounding and sequential intersection make forecasts nested; exhaustion is the center singleton, and $n = 0$ uses $v_h(0)(N_h - m)/N_h$. These are scheduling predictions, never inference bounds.

Start from every cell's longest forecast prefix; if incomplete, there is no feasible forecast plan. Otherwise visit cells by decreasing longest-prefix cost, then cell ID, and binary-search the smallest option preserving complete-request feasibility. This deadline-limited coordinate pass retains a checked feasible incumbent; it is coordinatewise minimal only when the pass finishes. The compact solver may replace it only with a cheaper, exactly checked forecast allocation. Actual decisions still require observed intervals. A retained allocation schedules

Algorithm 1 Shared-evidence executor (all allocation policies)

Compile contrasts; charge and observe pilot P.
Freeze cells, priors, risks and independent cell permutations.
Initialize round $t = 0$ and intervals $[0, N_h]$.
loop
  if a sticky failure occurs **then**
    Return error and submitted cost.
  end if
  Propagate signed intervals; check all decision certificates.
  if every decision is justified **then**
    Return values/sets, proofs and cost.
  end if
  if cap spent or no unread cell remains **then**
    Return unresolved items and cost.
  end if
  $t \leftarrow t + 1$.
  if t is divisible by four **then**
    Choose uniformly among nonexhausted cells (seeded).
  else if an unvisited cell remains **then**
    Choose largest N_h, breaking ties by largest cell ID.
  else
    Ask the allocator for a cell and batch size.
  end if
  Use 8 reads for a first visit, otherwise at most 16; Forecast may request fewer. Clip to unread count and remaining cap.
  Charge the full batch; acquire its next formal prefix.
  Update count intervals after each returned record.
end loop

the cell maximizing its unfulfilled planned reads divided by remaining population, then unfulfilled reads, then smallest cell ID. Its next batch is clipped to that allocation. Between replans it keeps this allocation; fallback uses it when available, otherwise DifficultyGreedy. The clipped/deduplicated grid is $\{0, 16, 32, 64, 128, N_h - n_h\}$, with a 50 ms solver limit, 150 ms soft planning limit, one thread and replanning after eight rounds or when no planned reads remain. This period was chosen among $1, 4, 8$ on four earlier airline development queries. Soft deadlines can change rerun actions; these are scheduling settings, not efficiency guarantees.

Algorithm 1 gives the common execution loop. Policies see observed states and frozen parameters; unread labels, identities and permutations are hidden. A forecast may exceed the remaining budget, but each submitted action is clipped to the cap. Completion is checked after a successful batch: all its labels are charged and processed even if an intermediate prefix would suffice. Failed/partial batches retain all charges, return error and suppress claims. A sticky inconsistency anywhere in that batch suppresses output. Rendering failures also preserve charges.

F. A Constructed Check on the Frontier's Scope

Consider a hidden all-zero population: three disjoint 100-record cohorts A, B, X, fourteen flags/record, and 28 strict comparisons $\mu_{i,m} > \mu_{X,m}$ $(i = A, B)$. One read reveals a whole vector. Uniform five-record pilots cost 15, leaving 95 per cohort; subsequent within-cohort draws are uniform. This mathematical common-right-anchor request is separate from MAST.

TABLE I
SEPARATE ALL-ZERO CONSTRUCTION: TOTAL VECTOR ACQUISITIONS, INCLUDING 15 PILOT RECORDS. ALL ROWS USE GLOBAL RISK 0.05. ONLY THE FIRST TWO COMPARE THE SAME STRICT REQUEST; THE LAST CHANGES BOTH THRESHOLD AND PREDICTOR. THESE ARE MATHEMATICAL COSTS, NOT MEASURED MAST RESULTS.

Proof family	Request	Acquisitions
Specified 42-stream family	$\theta > 0$	205 (exact)
OR count certificate	Same $\theta > 0$	≤ 197
Beta$(1, 1)$ count intervals	Changed: $\theta > 0.05$	≤ 171

The specified family retains history-feasible counts with each marginal wealth $e_h = q_h / p_{h,K_h} < 840$ and $\sum_h e_h < 840$, plus sound affine consequences ($42 = 3 \times 14$ streams). An unseen A/B positive stays feasible since $95 + 41 < 840$; both cohorts must be exhausted, costing 205.

A different OR-of-flags certificate uses $e_i(n) = \mathbf{1}\{\text{no positive OR seen}\}\, 95/(95 - n)$, $n < 95$, after zero pilots. Under the union positive-count null, $\min(e_A, e_B)$ is dominated by a true-null process. Threshold 20 is crossed after 91 reads per A/B, costing 197; positives trigger census fallback subject to the cap. Zero A/B rates settle all comparisons false regardless of X.

The changed threshold 0.05 with Beta$(1, 1)$ prediction and per-stream risk $1/840$ instead needs 78 zero reads per A/B, costing 171. The artifacts give full validity/fallback derivations and finite checks. At cap 200, these cases illustrate how proof family and task tolerance change feasibility.

G. Checking the Without-Replacement Assumption

Our kernel uses the predictable next-label null law in Section II-B. A scoped check explains this choice. The WOR transfer displayed by Ryu and Wornell [12, Sec. 3.2, Prop. 10] substitutes the final remaining mean into historical wealth. With counts $(3, 7)$, $N = 10$, KT prior $(1/2, 1/2)$ and risk $\delta = 1/20$, two initial category-one draws occur with probability $1/15 > \delta$. Their mixture likelihood is $3/8$; substitution of remaining mean $(1/8, 7/8)$ gives wealth $(3/8)/(1/8)^2 = 24 > 20$. The predictable-null ratio is instead $(3/8)/(1/15) = 45/8$. This fixed-time exclusion diagnoses that displayed transfer, leaving WR-KT and other vector results intact. The technical artifacts retain the full diagnostic.

IV. EXPERIMENTAL DESIGN

Primary saved-label replays compare allocation; the same-path oracle and endpoint intervention measure proof opportunities and signed overlap. Earlier-corpus replays test replication, MAST compares inference architectures using vector

acquisitions, and live queries measure forwards/tokens/time. The construction in Section III-F is mathematical.

A. Natural Overlap and Reference Predicates

The identity-enriched CivilComments test split in WILDS [13] supplies our primary population. A fixed salted metadata hash selects 30,000 rows having at least one of eight existing mention scores at least 0.5. Selection uses no target labels or outcomes. Duplicate texts retain distinct source IDs. Cohorts describe *content mentions*, not authors' identities or representative demographic groups. Cohort sizes are 2,011–9,409; all 28 intersections are nonempty (98–3,291 rows), and 8,767 rows (29.22%) have multiple memberships (Fig. 2). Existing toxicity, insult, identity-attack and obscene-language scores yield four crowd references, positive at decimal score ≥ 0.5, including ties.

Per predicate, the workload has 56 directed zero-margin comparisons, 28 unordered equivalence judgments (tolerance 0.05), 16 dominance summaries, 16 mixed summaries and 16 four-cohort rankings (7 top-1, 9 top-2). A dominance request asks three comparisons against one anchor; a mixed request asks one comparison and two equivalences across six distinct cohorts. Thus 116 nonranking and 16 ranking requests per predicate yield 464 primary and 64 secondary contracts. Ten seeds give 4,640 primary units and 1,280 three-decision summary units per reference/method. Truth and certifiability never filter these units.

Population, queries, thresholds, operator, original three policies and seeds were internally hash-frozen before target conversion and model inference. This is an internal freeze, not external preregistration. Ten independent whole-workload blocks supply separate pilot, formal-permutation, exploration and planning streams, shared across methods.

A complete $30{,}000 \times 4$ Qwen3-8B matrix [14] supplies the second reference on the same population. Its pinned operator fixes weights/tokenizer, prompts, BF16, SDPA, a nonthinking chat template, one record per forward and next-token $0/1$ scoring (ties positive), with no generation or KV/result reuse. All contexts fit 4,096 tokens without truncation. Operational canaries check stability separately. Allocation experiments replay this saved matrix: their charged label acquisitions are not fresh forwards. Matrix construction and live-query timing are separate studies. Both crowd and model references are fixed labels, not asserted semantic truth.

B. GoEmotions Source Validation

The three checksum-verified GoEmotions release files [15] contain 58,011 distinct IDs; 57,877 have at least two clear, distinct raters. A fixed salted metadata hash selects 30,000 eligible IDs without using emotion scores or outcomes. Four targets—anger, joy, sadness and fear—are positive when at least half the fixed clear raters select them, including ties.

Eight metadata-only cohorts use the four most frequent subreddits within the selected 30,000 IDs, length/time at least their medians on that same selected population, and question/exclamation marks. Their sizes are $139, 125, 122, 122, 15{,}070, 15{,}000, 3{,}044, 4{,}145$. The same 132 abstract cohort-index templates transfer to each emotion, with ten fresh four-stream seeds. Inputs and policy code were hash-frozen before target conversion, after CivilComments outcomes were known. This is a separately sourced, crowd-only validation using the same request design; it introduces no new model or latency measurement.

C. Common Execution and Comparators

Here I_q denotes request q's effective support. The cap is $B = \lceil 0.5|I_q| \rceil$, request risk 0.05, and charged pilot size $\min\{256, \lfloor |I_q|/10 \rfloor, \lfloor B/4 \rfloor\}$. Label-free TF–IDF features (up to 10,000 uni/bigrams, minimum frequency two, sublinear TF) use SVD-128/L2 normalization. Pilot-only logistic regression uses $C = 1$ and at most 1,000 iterations. For an empty/single-class pilot or a fit/score failure, all residual scores become $(s_p + 0.5)/(n_p + 1)$; the fallback is recorded. Average the frozen scores within cell h to $\bar{p}_h$, and set its proxy-Beta shapes to $1 + \lfloor 20\bar{p}_h \rfloor$ and $21 - \lfloor 20\bar{p}_h \rfloor$. There is one cell per coefficient signature, with this frozen prior. All methods share it, eight-record cell initialization, later batches of sixteen, every-fourth-round exploration and the exact inference/stopping loop. Only cell allocation changes.

Matched chooses the nonexhausted cell with fewest formal records read, n_h, excluding the pilot; ties use smallest cell ID. *Forecast* uses Section III-E.

DifficultyGreedy prioritizes predicted progress toward an easily completed proof. It is an internal heuristic, distinct from Adaptive Dual Greedy [5]. Its defining score uses the original contrast units: let $[l_j^\theta, u_j^\theta]$ be the original contrast interval, before threshold shifting and integer scaling, and b_j its threshold. For atom j, let $W_j = u_j^\theta - l_j^\theta$, $m_j = (l_j^\theta + u_j^\theta)/2$ and $e_j = \sigma_j(m_j - b_j)$ for its original threshold b_j, where $\sigma_j = 1$ for a lower atom and -1 for an upper atom. Set

$$r_j^{\pm} = \frac{W_j}{2\max\{\pm e_j, \varepsilon\}}, \qquad \mathrm{LME}(x) = \log\left(\frac{1}{|x|}\sum_i e^{x_i}\right). \tag{16}$$

For an unresolved decision, let $\mathcal{V}_d$ be its positive witnesses not yet hit by a false atom, and R_w their unresolved atoms. Define

$$P_d = \min_{w \in \mathcal{V}_d} \mathrm{LME}(r_j^+ : j \in R_w), \tag{17}$$

$$F_d = \mathrm{LME}\left(\min_{j \in R_w} r_j^- : w \in \mathcal{V}_d\right). \tag{18}$$

The objective sums $\min(P_d, F_d)$ for binary decisions and P_d for rankings. Resolved decisions contribute zero. A blocked strict ranking with no eligible positive witness also contributes constant zero but stays incomplete. Here $\varepsilon = 10^{-4}$ and LME temperature is one; stable log-sum-exp evaluates it.

For an active cell ($n_h < N_h$), candidate batch size is $b = \min\{16, N_h - n_h, B - C_{\mathrm{spent}}\}$. Here C_{spent} is submitted calls so far. The batch predicts width reduction Δ_h using the finite-population square-root shrinkage of Section III-E (no shrinkage at $n_h = 0$; zero residual width at exhaustion).

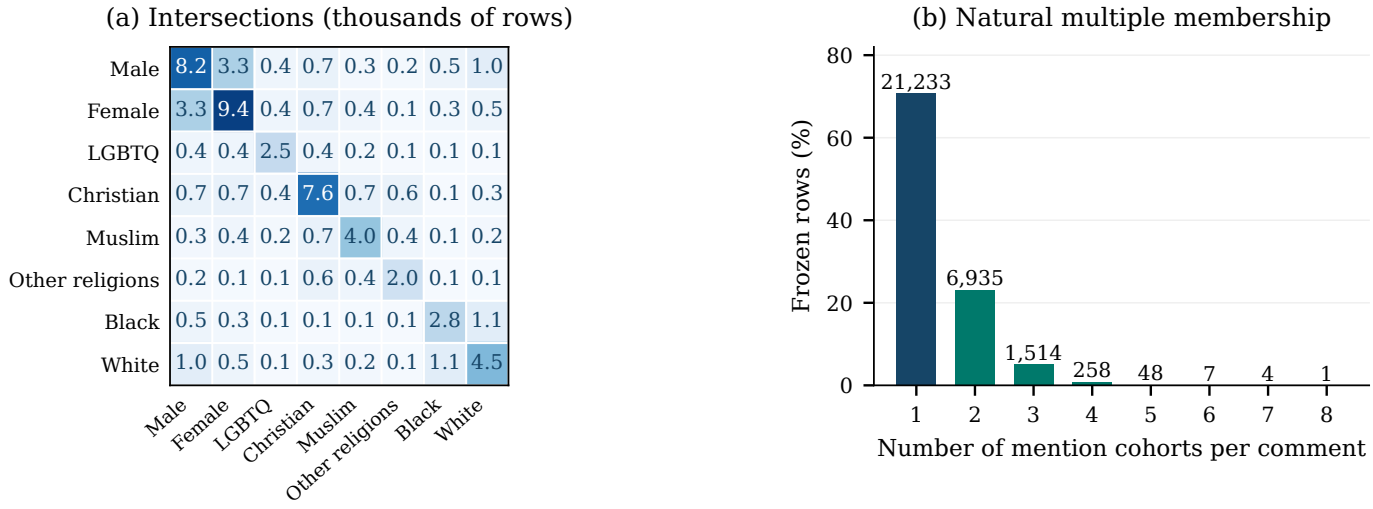


Fig. 2. Metadata of the 30,000 hash-selected eligible CivilComments rows. Cohorts use existing content-mention scores ≥ 0.5. In (a), diagonal entries are cohort sizes and off-diagonal entries are intersections, rounded to thousands; exact counts accompany the artifacts. Panel (b) counts each row once. Neither panel uses target labels.

Let c_{jh} be the original, unscaled contrast coefficient. Replace each W_j by $\max\{0, W_j - |c_{jh}|\Delta_h\}$, holding midpoints, unresolved atoms and witness eligibility fixed. Choose maximum predicted objective reduction divided by b; unit costs are one. Gain/cost ties within $10^{-12}\max\{1, |\text{best}|\}$ use smallest cell ID. Nonpositive gain falls back to largest residual-cell size N_h, then smallest ID. The shared checker alone certifies completion.

WidthGreedy targets normalized interval reduction. It prunes settled decisions/atoms and refuted positive witnesses, then deduplicates sign/scale-equivalent live contrasts. For live directions $\mathcal{C}$, normalize $\bar{c}_{jh} = c_{jh}/R_j$ by the fixed post-pilot range $R_j = \sum_h N_h|c_{jh}|$. Its batch score is the mean predicted normalized width reduction, $|\mathcal{C}|^{-1}\sum_{j\in\mathcal{C}}|\bar{c}_{jh}|\Delta_h/b$. *VarianceGreedy* targets empirical variance reduction with the same pruning/normalization, scoring

$$\frac{1}{|\mathcal{C}|b}\sum_{j\in\mathcal{C}}\bar{c}_{jh}^2 N_h^2 \widehat{S}_h^2\left(\frac{1}{n_h} - \frac{1}{n_h + b}\right), \tag{19}$$

where $\widehat{p}_h = (s_h + 0.5)/(n_h + 1)$ and $\widehat{S}_h^2 = N_h\widehat{p}_h(1 - \widehat{p}_h)/(N_h - 1)$ for $N_h > 1$. Only revealed formal counts are used. Common initialization normally supplies $n_h > 0$; otherwise relevant unobserved cells are explicitly prioritized. Both controls use relative score ties of $10^{-12}|\text{best}|$, then cell ID, and the same largest-N_h nonpositive-gain fallback. This adapts Neyman variance allocation [16].

For one unresolved scalar comparison, DifficultyGreedy's score reduces to $W/[2\max\{|m - b_0|, \varepsilon\}]$. Its predicted gain per batch cost is therefore a common positive factor times $|c_h|\Delta_h/b$. This is exactly WidthGreedy's candidate ordering in real arithmetic, even with clipped batch sizes. The threshold affects stopping but cancels from this allocation. Floating ties can differ. Single-contrast equivalence also preserves width ordering for equal-price candidates; no general unequal-price equivalence is claimed. The scalar case thus cannot establish a distinctive Boolean allocation benefit.

MaxDifficulty isolates cross-decision gain aggregation on all 128 three-decision summaries. It replaces the sum of decision-level DifficultyGreedy marginal gains by their maximum, retaining the shared evidence, checker and every other execution setting. It is not independent label acquisition. The stronger controls add 23,680 CivilComments runs: WidthGreedy/VarianceGreedy on every contract and MaxDifficulty on summaries, both sources and all seeds. These post-hoc controls were code-frozen before execution. GoEmotions uses the same controls with DifficultyGreedy: 17,120 planned runs, including summary-only MaxDifficulty; no Forecast tuning is transferred.

For CivilComments rankings, *Boundary* uses relative estimates $\widehat{\theta}_g = c_{g,o}^P + \sum_h c_{(g,o),h}N_h s_h/n_h$, with lexicographically first origin o, exact pilot difference $c_{g,o}^P$ and $\widehat{\theta}_o = 0$. It selects the top-k estimates (cohort-ID ties), then the inside cohort with smallest relative lower bound and outside cohort with largest relative upper bound, targeting their direct width reduction. It is a scoped boundary adaptation. The total is $31{,}680 + 1{,}280 + 23{,}680 + 17{,}120 = 73{,}760$ planned main-control replays; old-source, MAST and live studies are separate. Crowd/model sources are views of one population; GoEmotions is separately sourced.

D. A Fixed Released-Judge Vector Request

Each MAST-Data record [17] is a released multi-agent task trace with judge flags. Three fixed cohorts contain 100 records each: MetaGPT/GPT4o (A), MetaGPT/Claude (B), and ChatDev/GPT4o (C). Unlike the primary scalar calls, one acquisition here charges a single record and reveals its 14 released binary judge flags. Completion requires all 28 values $\mu_{A,j} > \mu_{B,j}$ and $\mu_{A,j} > \mu_{C,j}$, with ties false. Targets are

released judgments, without human-gold or original-model-version claims.

Seven programs each retain eight paths. The budgets $75, 150, 225, 300$ are readouts of the same planned execution paths, rather than fresh runs at each budget. Precommitted permutations supply local pilot/residual orders, joint full-population orders and the inputs to native seeded shuffles. Prefixes across architectures may differ.

a) Programs.: Local–RR, Width and DWidth share count confidence sequences and a complete-output checker, selecting by round robin, summed width reduction or difficulty-weighted width reduction. Five charged pilot vectors/cohort set the residual-95 mixture $\frac{1}{2}\,\mathrm{Beta}(1,1)+\frac{1}{2}\,\mathrm{Beta}(1+s_p, 6-s_p)$; projection retains pilot positives s_p and denominator 100. Each of 42 streams receives risk $1/840$ (total 0.05).

NativeExact uses Evergreen's native seeded plans and exact cohort aggregates [4]. NativeProgressive–RR uses synchronized prefixes, full-100 inference, recorded seeds, every-prefix evaluation, grid 100, outward integer readout and sticky intersection. These are configured adapters with audited floating risks and fixed scheduling.

Joint–RR and Joint–LogGreedy share predictable without-replacement Kelly bets and verified witnesses for an entire finite maximal upper-null cover [9]. Incomplete optimization is unresolved. Classification risk is $1/560$/comparison, shared by its two signs; correlated-flag capitals are not multiplied. RR cycles cohorts; LogGreedy sums predicted improvements in clipped log-evidence bottlenecks using fixed past-only Beta predictions.

b) Accounting and endpoint.: Vector attempts, including pilots and failed/unfinished tails, are charged before return. Paid prefetch is distinguished from committed prefixes; semantic caches start empty. Credit requires resource eligibility, verified normal durable closure, exact cost $\le B$, all 28 correct supported online outputs and no earlier source-count exclusion or erased inference error. All eight paths remain per program. Censoring means that an execution was interrupted before verified normal closure; it receives no credit. All four censors, including both Native startup censors, remain permanent. Unknown cost tails stay unknown; paths are not retried.

Exact census remains valid. PEAK/target intervals [8], [10] and native repeated-draw accounting are outside this configured comparison. Vector acquisitions are not forwards, tokens or monetary costs.

E. Metrics, Audits and Uncertainty

Primary completion retains all nonranking query–seed units and requires every answer and selected proof atom to match its full reference. Correct vectors with source-invalid witnesses are separate. Wrong positive/negative claims, unresolved requests, sticky inference errors and framework errors remain. Primary costs average submitted labels/support over all planned units; a sensitivity charges errors and invalid proofs to their full caps. Jointly complete paired savings are secondary, with their selection denominator.

Balanced aggregation weights four predicates and three families equally: pairwise, equivalence and summaries, the last pooling equal-sized dominance and mixed groups. Paired 95% percentile intervals use 20,000 resamples of ten whole-workload seed blocks, conditional on source and workload. Shared records, cohorts, reversed directions and templates make run records dependent. Soft-deadline nondeterminism is not separately isolated. These intervals establish neither request-population nor corpus generalization, and do not apply to the eight-path MAST replay.

An executor/compiler-independent analyzer reconstructs fixed-reference means and Boolean families, checks proof sides/costs/fingerprints, and scans available CS paths. Preselected and outcome-selected diagnostics retain separate denominators. Source exclusion, wrong assertion, sticky failure and framework error are audited separately.

F. Oracle, Replications and Runtime Scope

The oracle selects 48 nonranking IDs by support strata/salted hashes: four predicates, comparison/equivalence/dominance/mixed subtypes and three strata. Seeds $0, 1$ and both references yield $48 \times 2 \times 2 = 192$ cases of the original three policies, with explicit source-valid atom restrictions, full-path reconstruction and matched risk/kernel. Missing/unbounded cases stay. Verified $[L, U]$ total gaps include integrality and solver gaps; floating incumbents are not lower bounds. The reported relative gap is $(\overline{C} - \underline{C})/\max\{1, \overline{C}\}$, using total costs including pilot. Later numerical bound recovery is separately versioned and applies to all 192 selected cases.

The post-hoc endpoint intervention keeps all 192 cases/realized paths. Linked virtual appearances use saved signed grids, feasible prefixes and legal endpoints. Each receives 0.25 s coordinate work, three-second LP and three-second MIP limits, one solver thread. Independent exact arithmetic checks dual lower bounds, physical single-charge costs and endpoint proofs. No online policy is rerun or retuned. For bound-bearing cases, let a_i be policy calls on a correct completion and cap B_i otherwise, and $A = \sum_i a_i$. Capped hindsight headroom is

$$\left[\frac{A - \sum_i \min(\overline{C}_i, B_i)}{A},\quad \frac{A - \sum_i \min(\underline{C}_i, B_i)}{A}\right].$$

A proved absent source-valid proof contributes the cap at both endpoints; unavailable bounds retain separate status.

Earlier sources contain 40,000 CFPB narratives [18], [19] and 18,569 OpinRank reviews [20], four saved model predicates each. Their 480 contracts split $80/80/320$ train/development/test, with disjoint test cohorts. All 320 test contracts use the revised executor, three policies, ten seeds and 50% caps (9,600 replays). These post-hoc saved-label replications have an unpinned upstream model revision and no human validation of the eight predicates. Historical one-sided results remain separate.

The metadata-hash-selected live schedule contains five CivilComments queries and one airline ranking (six cases

total). The ranking uses the 14,640-row Twitter US Airline Sentiment source [21] (pinned-model negative sentiment). The warm matrix has six queries × three policies plus Boundary on two rankings (20 executions). Three fresh-process executions use the same CivilComments toxicity pairwise query, one per primary policy. Warm execution and fresh-process startup separately account physical forwards, tokens, pilot/proxy work, planning, inference, certification and rendering. Primary crowd outcomes were known; the schedule designer fixed metadata-hash selection before method-outcome inspection. Its results describe these six post-hoc cases and their warm/fresh-process conditions.

The public repository linked in the Introduction provides fixed CPU-study inputs, method/analysis sources, bounds and runtime receipts. Its manifests document external requirements, trace availability and MAST audit versions.

V. Results

A. Precision Controls and Whole-Request Completion

Table II retains all 4,640 primary units per reference/method. CivilComments DifficultyGreedy adds $36/54$ crowd/model completions over WidthGreedy: $0.78/1.16$ points, with paired whole-block intervals $[0.69, 0.86]/[0.91, 1.44]$. Relative reductions in mean all-unit calls/support are $0.54\%/0.60\%$. Against VarianceGreedy, gains are $3.25/5.41$ points and reductions $2.89\%/3.94\%$, narrowing the advantage over Matched ($5.51\%/6.04\%$ fewer calls). Shared coefficients, support filtering, labels and inference isolate allocation.

DifficultyGreedy and WidthGreedy agree on calls, status, completion, answer/proof correctness and errors across all 6,720 paired scalar/equivalence query–seed–reference units. This observable equality does not imply identical read order. Completion differences occur only in multi-judgment summaries, consistent with the scalar reduction. Figure 3 retains unsuccessful units.

On independently sourced GoEmotions, DifficultyGreedy completes $2{,}896/4{,}640$ primary units, WidthGreedy 2,894 and VarianceGreedy 2,772. The width comparison is a near tie: $+0.043$ completion points with interval $[-0.065, 0.129]$, and 0.20% fewer normalized calls. Against variance allocation the differences are $+2.67$ points and 2.98% fewer calls. Another 3,360 scalar/equivalence pairs agree on the checked observable fields. Summary gain aggregation completes 837 versus MaxDifficulty's $833/1{,}280$ units: $+0.31$ points with interval $[-0.08, 0.70]$, and 0.47% fewer calls. GoEmotions therefore does not reproduce a clear completion advantage over width or maximum-decision allocation. All 17,120 records reconcile with the independent reference/prefix/proof audit; fourteen sticky errors across primary and ranking controls remain, with no source-wrong proof. Secondary ranking completions are $317/312/256$ for difficulty/width/variance allocation.

Forecast beats Matched overall but loses $2.20/4.87$ CivilComments completion points to DifficultyGreedy and uses more calls. Its crowd summaries complete $695/1{,}280$ versus Matched's $432/1{,}280$, but pairwise/equivalence counts decline ($1{,}317/2{,}240$ versus $1{,}340/2{,}240$; $772/1{,}120$ versus $776/1{,}120$). Average gains need not hold per family. Balanced predicate/family aggregation preserves the main ordering.

TABLE II
ALL PRIMARY NONRANKING UNITS: 4,640 PER ROW. CIVIL C/M: CIVILCOMMENTS CROWD/MODEL; GO C: GOEMOTIONS CROWD. CORRECT REQUIRES EVERY VALUE AND SELECTED PROOF TO MATCH ITS REFERENCE. CALLS/$|I|$ AVERAGES ALL UNITS. E/W COUNTS EXECUTION ERRORS/SOURCE-WRONG PROOFS, RETAINED AS UNSUCCESSFUL.

Reference	Allocator	Correct	Calls/$\lvert I\rvert$	E/W
Civil C	DifficultyGreedy	2,886	0.3265	8/0
Civil C	WidthGreedy	2,850	0.3283	9/0
Civil C	VarianceGreedy	2,735	0.3363	10/0
Civil C	Matched	2,548	0.3456	28/0
Civil C	Forecast	2,784	0.3365	10/0
Civil M	DifficultyGreedy	3,384	0.3125	24/1
Civil M	WidthGreedy	3,330	0.3144	24/1
Civil M	VarianceGreedy	3,133	0.3253	22/0
Civil M	Matched	2,892	0.3326	37/1
Civil M	Forecast	3,158	0.3242	20/1
Go C	DifficultyGreedy	2,896	0.3084	3/0
Go C	WidthGreedy	2,894	0.3090	3/0
Go C	VarianceGreedy	2,772	0.3178	5/0

B. Gain Aggregation and Signed-Overlap Propagation

Holding shared labels and inference fixed, on all 1,280 summaries per CivilComments reference, summed versus maximum DifficultyGreedy gain completes $746/907$ versus $728/894$ crowd/model units: $1.41/1.02$ points, intervals $[0.94, 1.95]/[0.39, 1.72]$, and $0.84\%/1.29\%$ fewer normalized calls. Crowd gains are all dominance; model gains are eleven dominance and two mixed. This isolates gain aggregation.

All 192 endpoint-intervention cases reconcile with physical endpoints, single-charge reads and rational bounds. $\overline{C}_{\text{signed}} < \underline{C}_{\text{relaxed}}$ certifies extra necessary proof cost in $13/96$ crowd and $17/96$ model cases (Fig. 4); within these subsets, median lower bounds are $91/42$ extra total calls. Other cases are inconclusive. No pair proves a signed-feasible case crosses the 50% cap under relaxation. The relaxed study retains 191 finite brackets and one no-source-proof case; median gaps are $7.62\%/7.16\%$, maxima $54.40\%/61.37\%$. These bounds identify a proof-cost effect; online completion is not rerun.

C. What the Offline Frontier Establishes

All 192 preselected cases reconcile with all three methods' six initialization hashes, IDs, labels and per-cell CS endpoints. Within-cap source-valid atomwise proofs exist for $67/96$ crowd and $90/96$ model cases. Lower bounds exceed cap in $24/3$; another model case has no source-valid proof on any legal prefix. Five crowd/two model cases remain indeterminate. None is replaced or omitted.

DifficultyGreedy completes $62/96$ crowd and $82/96$ model; Forecast $61/75$; Matched $52/66$. Constructive within-cap proofs therefore remain uncompleted in 13 DifficultyGreedy and 21 Forecast cases across views. Many crowd noncompletions also face proved cost barriers. Figure 5 retains bound

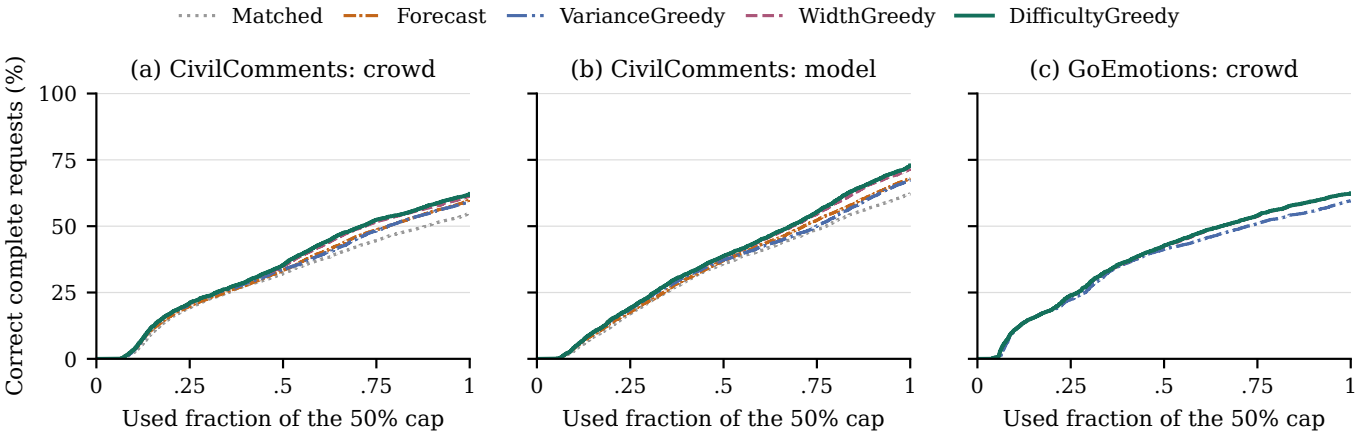


Fig. 3. All-unit reference-proof-correct stopping-cost distributions within the declared 50% cap, not restarted runs at each budget. CivilComments crowd/model share a population; GoEmotions is separately sourced. Incomplete, wrong-proof and error units never enter completed counts.

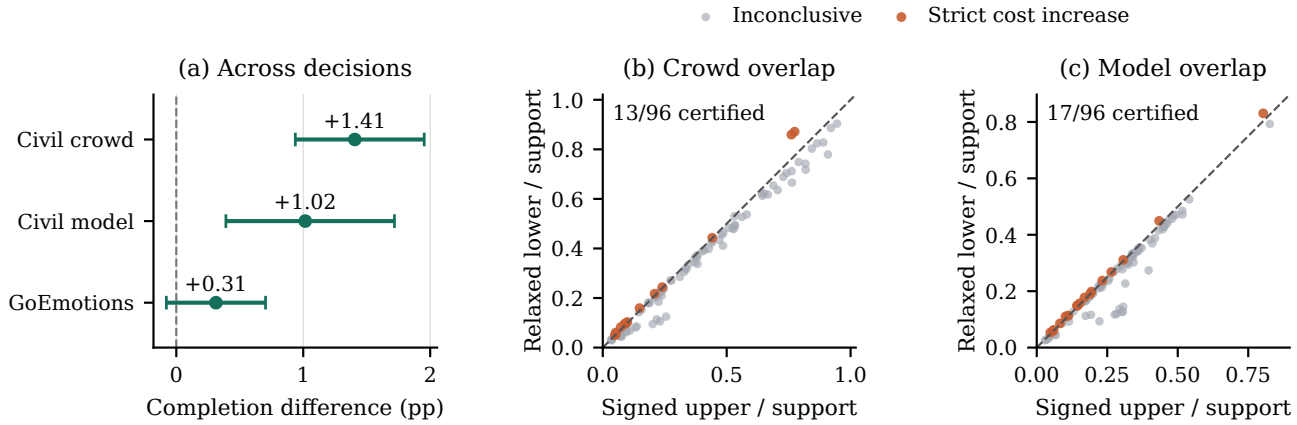


Fig. 4. Fixed-evidence mechanisms. (a) DifficultyGreedy (sum) minus MaxDifficulty (max) on 1,280 summaries/reference, with paired 95% seed-block intervals. (b,c) Signed upper versus relaxed lower total-cost bounds, normalized by support. Above equality proves extra necessary cost; gray points are inconclusive. All 192 planned cases remain; one has no source-valid proof. The cap is 0.5 support; no case proves a cap crossing.

uncertainty. Proof existence under hindsight witness/prefix choice does not establish an attainable causal executor improvement.

103/191 finite signed brackets reach 5% total-cost gap; the other 88 reach up to 63.14%. One case lacks a source-valid proof. A logged precision pass over all selected cases retains small rational dual multipliers without changing cases or upper proofs. DifficultyGreedy cap-charged summed-call headroom is 7.03–12.59% crowd/13.55–23.97% model: hindsight opportunity, not an online guarantee.

D. Optimization Work and Original-Corpus Replications

At 78,937 observed crowd planning states, MIP improves the forecast-feasible incumbent; this does not translate into better acquisition or isolate MIP's causal effect. Mean saved-label replay CPU is 1,272 ms, versus DifficultyGreedy 608 and Matched 531 ms, not live model time.

Post-hoc original-corpus replications retain 1,600 units/method/corpus. Matched/DifficultyGreedy/Forecast complete 653/721/701 CFPB and 228/326/298 OpinRank units, with mean normalized calls 0.3920/0.3805/0.3865 and 0.4838/0.4780/0.4800. All 9,600 records reconcile with independent source/prefix/cost/proof audits, including the source-invalid path reported below. The revised bidirectional all-unit results are separate from historical one-sided, certifiability-conditioned results.

A post-hoc diagnostic compares the first fresh feasible forecast's predicted total P (including pilot/earlier reads) with verified oracle lower bound L. $P < L$ proves optimism in 9/94 crowd and 6/91 model full-path source-covered cases. All 192 have plans; two crowd/five model source-excluding paths stay separate. Median covered-subset oracle gaps are 4.92%/4.98%. Predictions above loose lower bounds are inconclusive; this identifies neither cause nor a causal explanation of policy ordering.

Rankings are secondary. DifficultyGreedy/WidthGreedy/VarianceGreedy complete 353/324/312

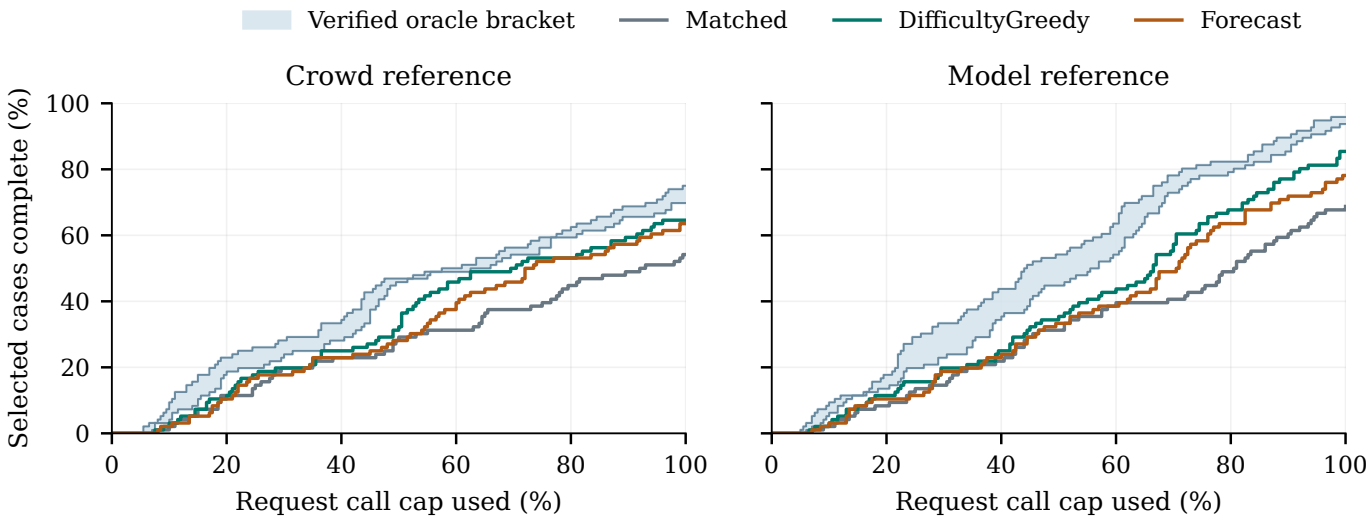


Fig. 5. Offline source-valid frontier, 96 planned cases/reference. At budget b, constructive upper costs count proved feasible completions; lower costs count cases not excluded. Their band is deterministic cost-bound uncertainty. Policy curves are stopping-cost distributions of fixed runs, retaining unsuccessful cases. The oracle uses future paths under the stated kernel, risks and proof rules.

TABLE III
PINNED MODEL LABELS RELATIVE TO EXISTING THRESHOLDED CROWD REFERENCES, ALL 30,000 ROWS PER PREDICATE. PERCENTAGES ARE DESCRIPTIVE; PRECISION/RECALL CONCERN THE POSITIVE CLASS.

Predicate	Agreement	Precision	Recall	Bal. acc.
Toxicity	83.03	48.16	36.79	64.48
Insult	74.86	19.65	70.81	73.01
Identity attack	74.39	17.25	69.46	72.11
Obscene	96.58	9.30	48.72	72.81

of 640 crowd and 454/422/392 model units; Forecast/Matched/Boundary counts are 342/298/295 and 431/338/355. These larger differences neither replace the primary endpoint nor establish superiority over canonical LUCB, IFocus or a semantic engine.

E. Sampling Failures and Semantic Reference Changes

Original three-arm CivilComments grids retain 50 crowd/100 model sticky inference errors and three source-wrong model false-equivalence outputs from one shared query/seed path. New controls retain their errors/wrong proofs (Table II); no favorable retry is taken. Another shared CFPB path has wrong proofs and values. Exact-rational checks permit the CivilComments and CFPB exclusions at nominal thresholds, without identifying a numerical defect. Noncoverage, sticky failure, unresolved output and semantic disagreement stay distinct. Dependent paths do not calibrate empirical risk.

Across 30,000 rows/predicate, model/crowd agreement is 74.39–96.58%, positive precision 9.30–48.16% (Table III). Prompts paraphrase source concepts, not identical historical questionnaires. Canaries check basic behavior/stability, not semantic fidelity; rare-label agreement does not establish a good positive classifier. Complete confusion counts, prevalence, recall, balanced accuracy and mention-cohort breakdowns are reported.

Full-reference targets differ on 108/464 nonranking contracts and selected sets on 13/64 rankings. Model-relative intervals concern model labels. Crowd answer agreement can coexist with crowd-invalid witnesses. The original eight predicates remain without human-reference validation.

Of DifficultyGreedy's 3,838 model-proof-valid query–seed completions across all 528 contracts, 747 have a crowd-inconsistent conclusion or selected proof (a union count). Among 5,250 DifficultyGreedy model-reference query–seed runs with a recorded terminal interval collection, 2,027 have at least one crowd contrast outside its model-target interval. These alternate-reference disagreements are not extra model-target noncoverage and do not justify human-reference confidence coverage.

F. A Fixed Cross-Architecture Request

Table IV reports all seven configurations and eight paths for the fixed 28-comparison request, using the endpoint of Section IV-D. All configurations have zero credited whole-request completions at budgets 75, 150 and 225.

MAST's DWidth is distinct from primary DifficultyGreedy. Width/DWidth have identical counts and costs on all eight paths, showing no added difficulty-weighting benefit. Joint–LogGreedy's 7/8 conditional costs cannot establish unconditional savings over Joint–RR's 8/8.

A post-hoc import-context correction reverified unchanged trajectories. NativeProgressive credits 5/8; two normally closed paths have correct final vectors but earlier count exclusions. At $(N, n, s, K) = (100, 99, 19, 20)$, binary64 clipping/count readout yields upper 19 despite preceding $[19, 20]$.

A separate audit finds interface losses in 94 of 5,049 proper-prefix states ($N = 100$); directed enclosures remove these

TABLE IV
ONE FIXED 28-COMPARISON MAST REQUEST. $k_{300}/8$: CREDITED PATHS; COST: CHARGED ACQUISITIONS AMONG CREDITED PATHS ONLY (DASH: NONE). N/C: NORMAL/INTERRUPTED CLOSURE; $E/V?$: HISTORICAL COUNT-EXCLUSION/VERIFICATION-UNKNOWN FLAGS, WHICH OVERLAP CLOSURE STATUSES. ALL EIGHT PATHS REMAIN PER PROGRAM.

Program	$k_{300}/8$	Cost	N	C	E	$V?$
Local-CS / RR	7/8	298	7	1	0	1
Local-CS / Width	8/8	278–288	8	0	0	0
Local-CS / DWidth	8/8	278–288	8	0	0	0
NativeExact adapter	7/8	300	7	1	0	1
NativeProgressive / RR	5/8	300	7	1	2	1
Joint-Kelly / RR	8/8	298	8	0	0	0
Joint-Kelly / LogGreedy	7/8	278–290	7	1	0	1

enumerated losses. These counts are not an error-rate estimate. No CI or method trajectory was rerun; artifacts retain the arithmetic and original records.

G. Scoped Real-Forward Runtime

The 23 executions use one RTX 5090 (32 GB) and pinned Qwen3-8B: 69,799 verified corpus forwards, 13,302,613 tokens, plus 192 setup-canary forwards/17,888 tokens. Independent checks verify proofs and 63,911 CS updates; artifacts retain full receipts.

Correct completion is $17/20$ warm, $3/3$ process-cold. All three unresolved warm arms reach the insult-equivalence 2,766 cap. Matched/DifficultyGreedy/Forecast submit $21{,}903/18{,}125/21{,}039$ warm requests, with mean walls $94.78/78.68/91.90$ s over all warm responses, including unresolved ones. Boundary's two-ranking subset differs. Only five jointly correct pairs give summed Matched/DifficultyGreedy and Matched/Forecast wall ratios $1.2410/1.0356$. Timings are single realizations in three chronological windows.

Warm query walls total 1,703.034 s; enclosing lifecycle walls total 1,731.961 s, including setup, I/O and teardown. Matched/DifficultyGreedy/Forecast process-cold parent walls $66.254/64.455/68.708$ s versus query $40.153/37.687/40.300$ s include startup and exit. Process-cold permits warm filesystem caches; features are separate.

One exactly checked count exclusion recurs in three warm/three cold dependent paths. Final proofs remain source-valid ($20/23$ complete).

Six full-support materialized lookups need no new forwards. A complete controlled cold-build ledger is unavailable, so these timings do not establish the query volume needed to amortize materialization.

VI. RELATED WORK

Semantic queries. LOTUS, ZenDB, DocETL and Bond optimize model operators [1], [22]–[24]; ABAE, InQuest, Thalamus and SUPG study expensive-predicate approximation [2], [3], [25], [26]. Evergreen verifies aggregate/ordinal claims [4]; SemViews separates prior-label planning from fresh certification [27]. CohortCert connects complete comparative vectors to shared evidence and checked completion costs.

Acquisition and inference. Boolean evaluation [5] and decision diagrams [7] assume known distributions; query-guided resolution uses Boolean provenance [6]. Verification-error models guide re-verification [28]. PEAK pools stream capital [8]; stratified/target-aware inference handles composite objectives [9], [10]. Model confidence sets use joint regions and target-dependent intersections [29]; constrained selection projects risk rectangles onto optima [30]. Our frontier fixes one interval proof architecture.

IFocus, LUCB and best-arm/multiple-answer identification target unresolved orderings or top-set boundaries under their respective assumptions [31]–[34].

VII. DISCUSSION AND SCOPE

The frontier conditions on paths, partition, kernel, risks and atomwise rules. Hindsight feasibility does not itself supply a causal policy; different proof systems can change feasibility, as the construction illustrates. The measured scope is eight cohorts, shared primary templates, one MAST request and 23 live executions. Fixed-label, per-request coverage leaves semantic fidelity, drift and session-wide guarantees to additional assumptions. Verification-error models [28] and batching/materialization [24], [27], [35] address complementary deployment questions.

VIII. CONCLUSION

CohortCert connects complete comparative requests to shared evidence and checkable proof costs. Its compiler, completion program and verified bounds distinguish feasible acquisition opportunities from barriers within a fixed inference architecture. The experiments identify signed-overlap effects, delimit allocation benefits and separate forecast planning from execution, supporting reproducible diagnosis of when to improve allocation or change inference rules.

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